

Using Positive and Negative Correlations to Interpret Data

Do Monkey Flower Species Differ in Allocating Their Energy to Sexual Versus Asexual Reproduction? Over the course of its life span, a plant captures only a finite amount of resources and energy, which must be allocated to best meet the plant's individual requirements for maintenance, growth, defense, and reproduction. Researchers examined how five species of monkey flower (genus *Mimulus*) use their resources for sexual and asexual reproduction.

How the Experiment Was Done

After growing specimens of each species in separate pots in the open, the researchers determined averages for nectar volume, nectar concentration, seeds produced per flower, and the number of times the plants were visited by broad-tailed hummingbirds (*Selasphorus platycercus*, shown here). Using greenhouse-grown specimens, they determined the average number of rooted branches per gram fresh shoot weight for each of the species. The phrase *rooted branches* refers to asexual reproduction through horizontal shoots that develop roots.



Data from the Experiment

Species	Nectar Volume (μL)	Nectar Concentration (% weight of sucrose/total weight)	Seeds per Flower	Visits per Flower	Rooted Branches per Gram Shoot Weight
<i>M. rupestris</i>	4.93	16.6	2.2	0.22	0.673
<i>M. eastwoodiae</i>	4.94	19.8	25	0.74	0.488
<i>M. nelsonii</i>	20.25	17.1	102.5	1.08	0.139
<i>M. verbenaceus</i>	38.96	16.9	155.1	1.26	0.091
<i>M. cardinalis</i>	50.00	19.9	283.7	1.75	0.069

Data from S. Sutherland and R. K. Vickery, Jr., Trade-offs between sexual and asexual reproduction in the genus *Mimulus*, *Oecologia* 76:330–335 (1998).

become more favorable. In the **Scientific Skills Exercise**, you can use data to determine which species of monkey flower are mainly asexual reproducers and which are mainly sexual reproducers.

Although sexual reproduction involving two genetically different plants produces the most genetically diverse offspring, some plants, such as garden peas, usually self-fertilize. This process, called “selfing,” is a desirable attribute in some crop plants because it heightens the probability that every ovule will develop into a seed. In many angiosperm species, however, mechanisms have evolved that make it difficult or impossible for a flower to fertilize itself, as we’ll see next.

INTERPRET THE DATA

1. A correlation is a way to describe the relationship between two variables. In a positive correlation, as the values of one of the variables increase, the values of the second variable also increase. In a negative correlation, as the values of one of the variables increase, the values of the second variable decrease. Or there may be no correlation between two variables. If researchers know how two variables are correlated, they can make a prediction about one variable based on what they know about the other variable. **(a)** Which variable(s) is/are positively correlated with the volume of nectar production in this genus? **(b)** Which is/are negatively correlated? **(c)** Which show(s) no clear relationship?
2. **(a)** Which *Mimulus* species would you categorize as mainly asexual reproducers? Why? **(b)** Which species would you categorize as mainly sexual reproducers? Why?
3. **(a)** Which species would probably fare better in response to a pathogen that infects all *Mimulus* species? **(b)** Which species would fare better if a pathogen caused hummingbird populations to dwindle?

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Mechanisms That Prevent Self-Fertilization

The various mechanisms that prevent self-fertilization contribute to genetic variety by ensuring that the sperm and egg come from different parents. Some species of plants cannot self-fertilize because different individuals have either staminate flowers (lacking carpels) or carpellate flowers (lacking stamens) (**Figure 38.14a**). Other plants have flowers with functional stamens and carpels that mature at different times or are structurally arranged in such a way that it is unlikely that an animal pollinator could transfer pollen from an anther to a stigma of the same flower (**Figure 38.14b**). However, the most common anti-selfing mechanism in flowering plants is **self-incompatibility**, the ability of a plant to reject its